

NFL EVERGAME 360 Dashboard - Data Schema Architecture

Powered by Sentrais OS & NIN Forensics Methodology

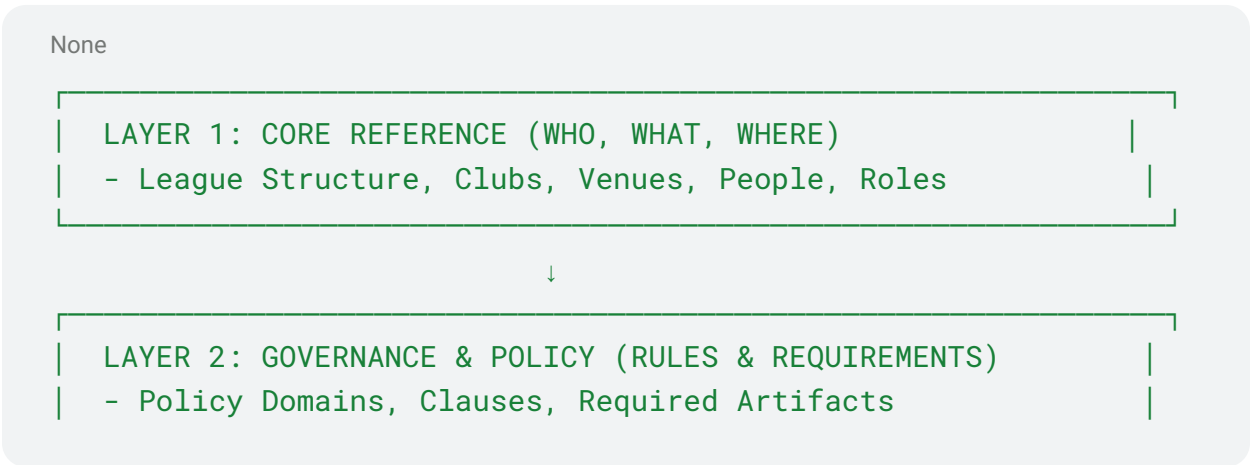
Version: 1.0
Date: November 15, 2025
Purpose: Foundation data model for unified NFL governance, compliance, and operational intelligence

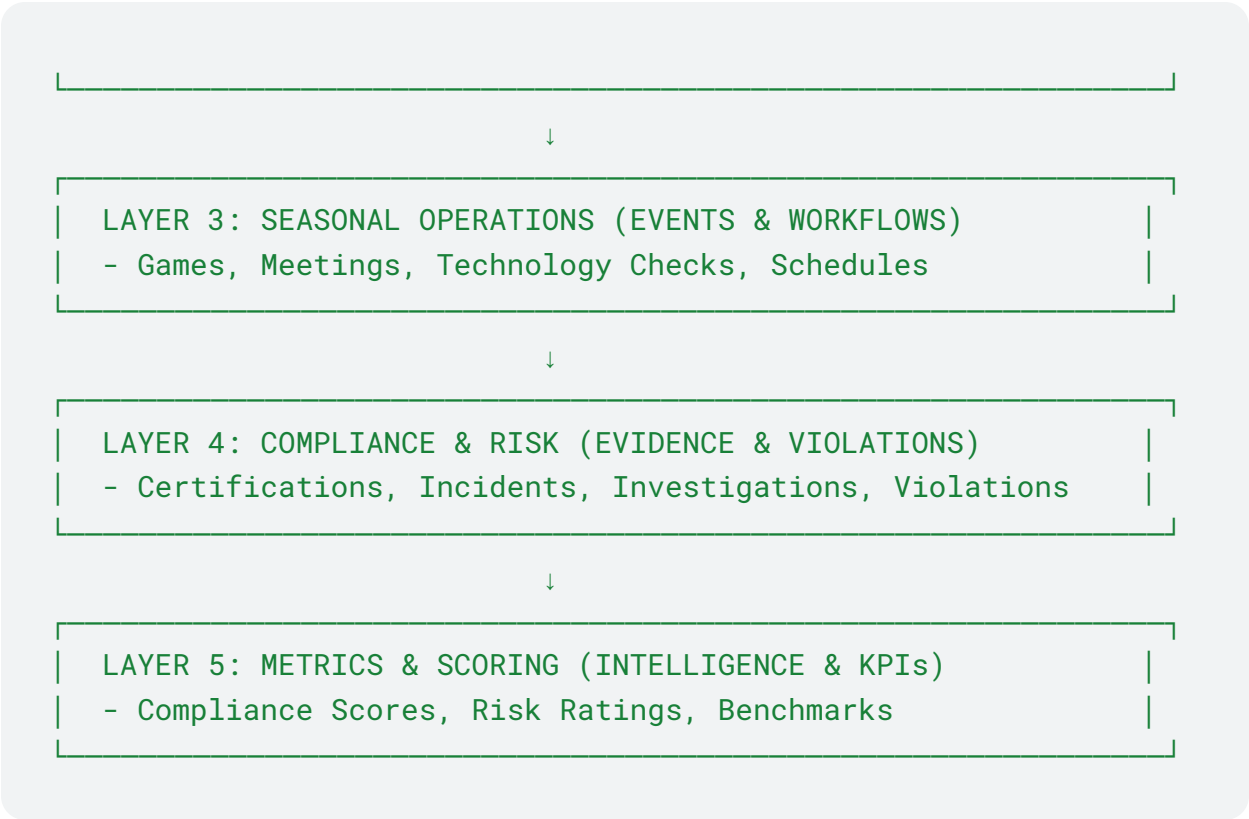
Schema Design Philosophy

The EVERGAME schema follows **NIN Forensics principles**:

- 1. **Entity Accountability:** Every entity has explicit responsibilities defined by policy
- 2. **Evidence Chain:** Every action requires timestamp, responsible party, and policy reference
- 3. **Temporal Awareness:** All compliance tracked across season phases
- 4. **Federated Access:** Role-based visibility (Commissioner ≠ Club GM ≠ Game Day Staff)
- 5. **Audit Immutability:** Evidence ledger cannot be altered, only appended

Architecture Overview





LAYER 1: CORE REFERENCE

1.1 season

Purpose: Represents a League year with defined phases and key dates.

Field	Type	Key	Description
season_id	UUID	PK	Unique identifier for NFL season
year	integer		Calendar year of regular season start (e.g., 2024)
label	string		Display label (e.g., "2024 Season")
status	enum		Planned, Active, Completed, Archived

start_date	date	Official League year start
end_date	date	Official League year end
created_at	timestamp	Record creation
updated_at	timestamp	Last modification

Indexes: year, status

1.2 season_phase

Purpose: Breaks season into governance phases with distinct compliance requirements.

Field	Type	Key	Description
phase_id	UUID	PK	Unique phase identifier
season_id	UUID	FK	Links to season.season_id
phase_type	enum		Offseason, Preseason, RegularSeason, Postseason, SpecialEvent
phase_name	string		Display name (e.g., "2024 Regular Season")
start_date	date		Phase start
end_date	date		Phase end
compliance_profile_id	UUID	FK	Links to compliance requirements for this phase
created_at	timestamp		Audit
updated_at	timestamp		Audit

Business Rules:

- Phases cannot overlap within same season
 - Compliance requirements change per phase (e.g., EMP updates due in Offseason)
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1.3 conference

Purpose: AFC/NFC organizational structure.

Field	Type	Key	Description
<code>conference_id</code>	UUID	PK	Unique identifier
<code>code</code>	string	UNIQUE	"AFC" or "NFC"
<code>name</code>	string		"American Football Conference"
<code>created_at</code>	timestamp		Audit
<code>updated_at</code>	timestamp		Audit

Static Data: Only 2 conferences exist (AFC, NFC)

1.4 division

Purpose: Four divisions per conference (North, South, East, West).

Field	Type	Key	Description
<code>division_id</code>	UUID	PK	Unique identifier
<code>conference_id</code>	UUID	FK	Links to <code>conference.conference_id</code>
<code>code</code>	string	UNIQUE	"AFC-NORTH", "NFC-WEST", etc.
<code>name</code>	string		"AFC North"
<code>created_at</code>	timestamp		Audit

`updated_at` timestamp Audit

Static Data: 8 total divisions (4 per conference)

1.5 club

Purpose: Core entity representing each of the 32 NFL franchises.

Field	Type	Key	Description
<code>club_id</code>	UUID	PK	Unique identifier
<code>division_id</code>	UUID	FK	Links to <code>division.division_id</code>
<code>code</code>	string	UNIQUE	Three-letter code ("KC", "SF", "BUF")
<code>name</code>	string		Official club name ("Chiefs")
<code>market</code>	string		Market ("Kansas City")
<code>full_name</code>	string		"Kansas City Chiefs"
<code>primary_venue_id</code>	UUID	FK	Default home stadium → <code>venue.venue_id</code>
<code>owner_name</code>	string		Principal owner
<code>status</code>	enum		Active, Inactive, Suspended
<code>founded_year</code>	integer		Year franchise established
<code>created_at</code>	timestamp		Audit
<code>updated_at</code>	timestamp		Audit

Indexes: `code`, `division_id`, `status`

Business Rules:

- A club must belong to exactly one division
- Club code must match NFL standard abbreviations

1.6 venue

Purpose: Stadiums, neutral sites, international locations.

Field	Type	Key	Description
venue_id	UUID	PK	Unique identifier
venue_name	string		"Arrowhead Stadium"
venue_type	enum		HomeStadium, NeutralSite, International, Practice
address	string		Street address
city	string		City
state_province	string		State/province (NULL for international)
country	string		Country code ("US", "GB", "DE", "BR")
capacity	integer		Seating capacity
surface_type	string		"Natural Grass", "FieldTurf", etc.
indoor_outdoor	enum		Indoor, Outdoor, Retractable
primary_club_id	UUID	FK	Home club (NULL for neutral sites)
latitude	decimal		GPS coordinates
longitude	decimal		GPS coordinates
status	enum		Active, Inactive, UnderConstruction
created_at	timestamp		Audit
updated_at	timestamp		Audit

Indexes: venue_type, country, status

1.7 person

Purpose: Individuals across League and clubs (employees, officials, contractors).

Field	Type	Key	Description
person_id	UUID	PK	Unique identifier
first_name	string		Legal first name
last_name	string		Legal last name
email	string	UNIQUE	Primary contact email
phone	string		Primary phone
organization_type	enum		LeagueOffice, Club, Vendor, Official, Contractor
organization_id	UUID	FK	Links to club or league department
employment_status	enum		Active, Inactive, Terminated, Retired
hire_date	date		Start date
termination_date	date		End date (NULL if active)
created_at	timestamp		Audit
updated_at	timestamp		Audit

Indexes: email, organization_type, employment_status

Privacy: PII fields encrypted at rest, access logged

1.8 role_definition

Purpose: Catalog of all required NFL roles (Game Day Designee, Tech Liaison, etc.).

Field	Type	Key	Description
<code>role_def_id</code>	UUID	PK	Unique identifier
<code>role_code</code>	string	UNIQUE	"GDD", "GDTL", "C2CT", "STADIUM_MGR"
<code>role_name</code>	string		"Game Day Designee"
<code>role_category</code>	enum		GameDay, Technology, Medical, Emergency, Broadcast, Compliance
<code>description</code>	text		Responsibilities per Game Operations Manual
<code>requires_certification</code>	boolean		TRUE if cert required
<code>certification_type_id</code>	UUID	FK	Links to <code>certification_type</code>
<code>recertification_frequency_days</code>	integer		Days until recert required (365 = annual)
<code>required_for_game_type</code>	enum[]		Array: Preseason, RegularSeason, Postseason, etc.
<code>policy_reference</code>	string		Manual section (e.g., "Section 2.B")
<code>created_at</code>	timestamp		Audit
<code>updated_at</code>	timestamp		Audit

Examples:

- Game Day Designee (GDD): Required for all games, annual certification
- Coach-to-Coach Technician (C2CT): Required for all games, tech cert
- Stadium Manager: Required for home games only

1.9 role_assignment

Purpose: Maps individuals to roles for a specific season/club/venue.

Field	Type	Key	Description
<code>assignment_id</code>	UUID	PK	Unique identifier
<code>person_id</code>	UUID	FK	Who is assigned → <code>person.person_id</code>
<code>role_def_id</code>	UUID	FK	What role → <code>role_definition.role_def_id</code>
<code>season_id</code>	UUID	FK	For which season → <code>season.season_id</code>
<code>club_id</code>	UUID	FK	For which club (NULL for League roles)
<code>venue_id</code>	UUID	FK	For which venue (NULL if not venue-specific)
<code>assignment_status</code>	enum		Active, Inactive, Pending, Expired
<code>effective_date</code>	date		When assignment became active
<code>expiration_date</code>	date		When assignment expires (NULL = no expiration)
<code>assigned_by_person_id</code>	UUID	FK	Who made the assignment
<code>created_at</code>	timestamp		Audit
<code>updated_at</code>	timestamp		Audit

Business Rules:

- One person can have multiple role assignments
- Assignments must align with employment status
- Expired assignments trigger compliance alerts

Indexes: `person_id`, `role_def_id`, `season_id`, `assignment_status`

1.10 `league_department`

Purpose: League Office organizational units.

Field	Type	Key	Description
department_id	UUID	PK	Unique identifier
department_name	string		"Football Operations", "Football Technology"
department_code	string	UNIQUE	"FOOTBALL_OPS", "TECH", "BROADCAST"
parent_department_id	UUID	FK	For hierarchical structure (NULL if top-level)
head_person_id	UUID	FK	Department head → person.person_id
status	enum		Active, Inactive
created_at	timestamp		Audit
updated_at	timestamp		Audit

Examples:

- Football Operations
- Football Technology
- Broadcast Operations
- Medical & Safety
- Security

LAYER 2: GOVERNANCE & POLICY

2.1 policy_domain

Purpose: High-level compliance categories per Game Operations Manual.

Field	Type	Key	Description
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domain_id	UUID	PK	Unique identifier
domain_code	string	UNIQUE	"INTEGRITY", "TECH", "MEDICAL_EMP", "INJURY_REPORT"
domain_name	string		"Integrity of the Game"
description	text		Overview of domain scope
owning_department_id	UUID	FK	League department responsible
severity_weight	decimal		1.0-10.0 (for compliance scoring)
created_at	timestamp		Audit
updated_at	timestamp		Audit

Standard Domains:

1. **INTEGRITY** - Integrity of the Game, Rule of Equity
2. **TECH** - Technology governance, C2C, IVR, tablets
3. **MEDICAL_EMP** - Emergency Management Plans, Exposure Control
4. **INJURY_REPORT** - Injury reporting timeliness and accuracy
5. **UNIFORM** - Uniform policy enforcement
6. **BROADCAST** - Media and broadcast compliance
7. **TICKETING** - Ticket policies (postseason)
8. **EMERGENCY** - Emergency response, unfair acts



2.2 policy_document

Purpose: Source documents (Game Operations Manual, Appendices, etc.).

Field	Type	Key	Description
document_id	UUID	PK	Unique identifier
document_code	string	UNIQUE	"GOM_2024", "APPENDIX_J", "INTEGRITY_POLICY"

document_title	string		"2024 Game Operations Manual"
document_type	enum		Manual, Appendix, Policy, Constitution, Bylaw
version	string		"2024.1.0"
effective_date	date		When document became active
supersedes_document_id	UUID	FK	Previous version (NULL if first)
status	enum		Draft, Active, Superseded, Archived
file_url	string		Link to PDF/source file
created_at	timestamp		Audit
updated_at	timestamp		Audit

2.3 policy_clause

Purpose: Specific enforceable rules from policy documents.

Field	Type	Key	Description
clause_id	UUID	PK	Unique identifier
policy_document_id	UUID	FK	Source document → <code>policy_document.document_id</code>
domain_id	UUID	FK	Policy domain → <code>policy_domain.domain_id</code>
clause_reference	string		"Section 2.B.3", "Appendix J.1.A"
clause_text	text		Exact language from manual
clause_summary	text		Plain-language description
enforcement_level	enum		Mandatory, Recommended, Guidance

<code>applies_to_entity_type</code>	enum[]	Array: Club, Venue, League, Individual
<code>violation_severity</code>	enum	Critical, High, Medium, Low
<code>standard_of_proof</code>	string	"Preponderance of evidence"
<code>created_at</code>	timestamp	Audit
<code>updated_at</code>	timestamp	Audit

Example:

- Clause Reference: "Appendix J.1"
- Summary: "All clubs must maintain an Emergency Management Plan"
- Enforcement: Mandatory
- Applies To: Club
- Severity: High

2.4 required_artifact

Purpose: Evidence/documents required by policy (EMP, certifications, etc.).

Field	Type	Key	Description
<code>artifact_id</code>	UUID	PK	Unique identifier
<code>clause_id</code>	UUID	FK	Policy requiring this → <code>policy_clause.clause_id</code>
<code>artifact_type</code>	enum		Certification, Plan, Report, Meeting, Inspection, Checklist
<code>artifact_name</code>	string		"Emergency Management Plan"
<code>description</code>	text		What must be included
<code>required_for_entity_type</code>	enum		Club, Venue, Individual, Game

frequency	enum		OneTime, Annual, Seasonal, PerGame, AsNeeded
submission_deadline_type	string		"30 days before season", "90 min before kickoff"
responsible_role_def_id	UUID	FK	Who submits → <code>role_definition.role_def_id</code>
approval_required	boolean		TRUE if League must approve
approving_department_id	UUID	FK	Which department approves
created_at	timestamp		Audit
updated_at	timestamp		Audit

Examples:

- EMP (Annual, Club-level, requires League approval)
- Pregame Technology Checklist (Per-Game, Game Day Tech Liaison)
- Injury Report (Per-Game, Medical staff, published to media)

2.5 certification_type

Purpose: Types of certifications required for roles.

Field	Type	Key	Description
cert_type_id	UUID	PK	Unique identifier
cert_code	string	UNIQUE	"INTEGRITY_CERT", "GAME_OPS_CERT", "TECH_CERT"
cert_name	string		"Integrity of the Game Certification"
description	text		What the cert covers
domain_id	UUID	FK	Related policy domain

validity_period_days	integer	365 for annual
issuing_authority	string	"NFL Football Operations"
requires_exam	boolean	TRUE if test required
passing_score	decimal	Minimum score (NULL if no exam)
created_at	timestamp	Audit
updated_at	timestamp	Audit

LAYER 3: SEASONAL OPERATIONS

3.1 game

Purpose: Individual game events across all types.

Field	Type	Key	Description
game_id	UUID	PK	Unique identifier
season_id	UUID	FK	Links to <code>season.season_id</code>
game_type	enum		Preseason, RegularSeason, WildCard, Divisional, Conference, SuperBowl, ProBowl, International
game_number	integer		Sequential number within season
game_date	date		Scheduled date
kickoff_time	time		Scheduled kickoff (local time)
timezone	string		Venue timezone
home_club_id	UUID	FK	Home team → <code>club.club_id</code>
away_club_id	UUID	FK	Away team → <code>club.club_id</code>

venue_id	UUID	FK	Where played → venue.venue_id
game_status	enum		Scheduled, InProgress, Completed, Postponed, Cancelled
broadcast_network	string		"CBS", "FOX", "NBC", "ESPN", "Prime Video"
week_number	integer		Week of season (1-18 for regular season)
playoff_round	enum		NULL, WildCard, Divisional, Conference, SuperBowl
weather_conditions	string		Conditions at kickoff
attendance	integer		Actual attendance
created_at	timestamp		Audit
updated_at	timestamp		Audit

Indexes: season_id, game_type, game_date, home_club_id, away_club_id

3.2 game_role_assignment

Purpose: Who is responsible for what during a specific game.

Field	Type	Key	Description
game_role_id	UUID	PK	Unique identifier
game_id	UUID	FK	Links to game.game_id
role_def_id	UUID	FK	Which role → role_definition.role_def_id
person_id	UUID	FK	Who is assigned → person.person_id
club_id	UUID	FK	For which club (home/away)

assignment_status	enum		Confirmed, Pending, Replaced, Absent
confirmation_timestamp	timestamp		When confirmed by club
confirmed_by_person_id	UUID	FK	Who confirmed
created_at	timestamp		Audit
updated_at	timestamp		Audit

Business Rules:

- Each required role must be assigned before game
- Missing assignments trigger compliance alerts

3.3 pregame_meeting

Purpose: Required meetings per Game Operations Manual.

Field	Type	Key	Description
meeting_id	UUID	PK	Unique identifier
game_id	UUID	FK	Links to <code>game.game_id</code>
meeting_type	enum		OfficialBroadcast, EmergencyPrep, NetworkSideline, Other
meeting_name	string		"90-Minute Emergency Preparedness Meeting"
required_by_clause_id	UUID	FK	Policy requirement → <code>policy_clause.clause_id</code>
scheduled_time_offset_minutes	integer		-120 means "120 min before kickoff"

<code>scheduled_datetime</code>	timestamp	Calculated scheduled time
<code>actual_start_time</code>	timestamp	When meeting actually started
<code>actual_end_time</code>	timestamp	When meeting ended
<code>meeting_status</code>	enum	Scheduled, InProgress, Completed, Cancelled, Late
<code>deviation_minutes</code>	integer	How late/early (NULL if on time)
<code>deviation_reason</code>	text	Explanation for deviation
<code>location</code>	string	"Home Team Locker Room"
<code>created_at</code>	timestamp	Audit
<code>updated_at</code>	timestamp	Audit

3.4 pregame_meeting_participant

Purpose: Who attended each pregame meeting.

Field	Type	Key	Description
<code>participant_id</code>	UUID	PK	Unique identifier
<code>meeting_id</code>	UUID	FK	Links to <code>pregame_meeting.meeting_id</code>
<code>person_id</code>	UUID	FK	Attendee → <code>person.person_id</code>
<code>role_def_id</code>	UUID	FK	In what capacity
<code>attendance_status</code>	enum		Present, Absent, Excused, Late
<code>check_in_time</code>	timestamp		When they arrived

check_out_time	timestamp	When they left
notes	text	Any relevant notes
created_at	timestamp	Audit
updated_at	timestamp	Audit

3.5 technology_system_check

Purpose: Validation of game day technology (C2C, tablets, IVR, etc.).

Field	Type	Key	Description
check_id	UUID	PK	Unique identifier
game_id	UUID	FK	Links to <code>game.game_id</code>
system_type	enum		CoachToCoach, AnalyticsTablets, InjuryVideoReview, RadioHeadsets, FrequencyCoord
check_type	enum		Setup, Validation, InGame, PostGame
required_by_clause_id	UUID	FK	Policy requirement
scheduled_check_time	timestamp		When check should occur
actual_check_time	timestamp		When check actually occurred
check_status	enum		Passed, Failed, PartialPass, NotPerformed
responsible_person_id	UUID	FK	Tech Liaison or FTGDA
failure_details	text		If failed, what went wrong
remediation_action	text		How issue was resolved

remediation_timestamp	timestamp	When resolved
created_at	timestamp	Audit
updated_at	timestamp	Audit

Business Rules:

- Failed checks must have remediation before kickoff
- Unresolved tech failures escalated to League Office

3.6 schedule_key_date

Purpose: Important League-wide dates (from Appendix H).

Field	Type	Key	Description
key_date_id	UUID	PK	Unique identifier
season_id	UUID	FK	Links to <code>season.season_id</code>
date	date		The key date
event_name	string		"EMP Submission Deadline"
event_type	enum		Deadline, MilestoneEvent, PhaseTransition
applies_to_entity_type	enum		AllClubs, LeagueOffice, Specific clubs
policy_reference	string		Manual section
description	text		What happens on this date
reminder_days_before	integer		Send alerts X days prior
created_at	timestamp		Audit

updated_at

timestamp

Audit

LAYER 4: COMPLIANCE & RISK

4.1 certification_record

Purpose: Tracks individual certifications (Integrity, Game Ops, Tech, etc.).

Field	Type	Key	Description
cert_record_id	UUID	PK	Unique identifier
person_id	UUID	FK	Who was certified → <code>person.person_id</code>
cert_type_id	UUID	FK	Type of cert → <code>certification_type.cert_type_id</code>
season_id	UUID	FK	For which season
issue_date	date		When cert was issued
expiration_date	date		When cert expires
cert_status	enum		Active, Expired, Revoked, Pending
exam_score	decimal		Score if exam required (NULL otherwise)
issued_by_department_id	UUID	FK	Issuing authority
revocation_reason	text		If revoked, why
created_at	timestamp		Audit
updated_at	timestamp		Audit

Business Rules:

- Expired certs trigger alerts 30/60/90 days before expiration

- Role assignments cannot be active without valid cert

Indexes: `person_id`, `cert_type_id`, `cert_status`, `expiration_date`

4.2 emergency_management_plan

Purpose: Club EMP submissions (Appendix J).

Field	Type	Key	Description
<code>emp_id</code>	UUID	PK	Unique identifier
<code>club_id</code>	UUID	FK	Owning club → <code>club.club_id</code>
<code>season_id</code>	UUID	FK	For which season
<code>version</code>	string		"2024.1.0"
<code>submission_date</code>	date		When submitted to League
<code>approval_date</code>	date		When approved (NULL if pending)
<code>approval_status</code>	enum		Pending, Approved, RejectedNeedsRevision, Expired
<code>approved_by_person_id</code>	UUID	FK	League approver
<code>basic_plan_complete</code>	boolean		Required component
<code>functional_annexes_complete</code>	boolean		Required component
<code>hazard_appendices_complete</code>	boolean		Required component
<code>completeness_score</code>	decimal		0-100% based on components
<code>mou_local_authorities</code>	boolean		MOU with local gov signed
<code>last_drill_date</code>	date		Most recent emergency exercise
<code>next_review_due_date</code>	date		When update required

file_url	string	Link to EMP document
notes	text	Reviewer feedback
created_at	timestamp	Audit
updated_at	timestamp	Audit

Business Rules:

- EMP must be updated annually during Offseason phase
- Incomplete EMPs trigger compliance alerts

4.3 exposure_control_plan

Purpose: Appendix K exposure control planning.

Field	Type	Key	Description
ecp_id	UUID	PK	Unique identifier
club_id	UUID	FK	Owning club → club.club_id
season_id	UUID	FK	For which season
version	string	"2024.1.0"	
approval_date	date		When approved
approval_statuses	enum		Approved, Pending, NeedsRevision
next_review_date	date		When update due
file_url	string		Link to plan
created_at	timestamp		Audit
updated_at	timestamp		Audit

4.4 injury_report

Purpose: Player injury disclosures per Injury Report Policy.

Field	Type	Key	Description
<code>injury_report_id</code>	UUID	PK	Unique identifier
<code>game_id</code>	UUID	FK	Related game → <code>game.game_id</code>
<code>club_id</code>	UUID	FK	Reporting club → <code>club.club_id</code>
<code>player_name</code>	string		Player name (anonymized for schema demo)
<code>injury_type</code>	string		"Ankle", "Knee", "Shoulder" (specific body part)
<code>injury_status</code>	enum		Questionable, Doubtful, Out
<code>occurrence_datetime</code>	timestamp		When injury occurred
<code>required_report_deadline</code>	timestamp		Policy deadline for disclosure
<code>actual_report_timestamp</code>	timestamp		When actually reported
<code>timeliness_status</code>	enum		OnTime, Late, VeryLate
<code>delay_minutes</code>	integer		Minutes late (NULL if on time)
<code>public_announcement_timestamp</code>	timestamp		When disclosed to media/fans
<code>announcement_method</code>	enum		PressRelease, Broadcast, Social, Multiple
<code>specificity_score</code>	decimal		0-100% (specific body part = 100%)
<code>subsequent_updates_count</code>	integer		How many status changes
<code>responsible_person_id</code>	UUID	FK	Medical staff who reported

notes	text	Additional context
created_at	timestamp	Audit
updated_at	timestamp	Audit

Business Rules:

- Late reports flagged for integrity review
- Vague announcements ("injury", "illness") trigger low specificity scores

4.5 uniform_violation

Purpose: On-field uniform policy violations.

Field	Type	Key	Description
violation_id	UUID	PK	Unique identifier
game_id	UUID	FK	Where occurred → <code>game.game_id</code>
club_id	UUID	FK	Which club → <code>club.club_id</code>
player_name	string		Player name
violation_type	enum		PantLength, UnapprovedGear, IncorrectColors, Other
detected_by_role	enum		Official, UniformInspector, LeagueObserver
detected_by_person_id	UUID	FK	Who detected
detection_time	timestamp		When during game
action_taken	enum		WarningIssued, PlayerRemoved, NoAction
corrected	boolean		Was violation fixed

correction_time	timestamp	When corrected
repeat_offender	boolean	Flagged if prior violations
fine_amount	decimal	Club or player fine
notes	text	Details
created_at	timestamp	Audit
updated_at	timestamp	Audit

4.6 integrity_case

Purpose: Investigations under Integrity of the Game Policy.

Field	Type	Key	Description
case_id	UUID	PK	Unique identifier
case_number	string	UNIQUE	"2024-INT-0042"
domain_id	UUID	FK	Policy domain → <code>policy_domain.domain_id</code>
clause_id	UUID	FK	Violated clause → <code>policy_clause.clause_id</code>
allegation_date	date		When allegation made
reporting_club_id	UUID	FK	Who reported (NULL if League-initiated)
subject_club_id	UUID	FK	Club under investigation
subject_person_id	UUID	FK	Individual under investigation (if applicable)
case_status	enum		AllegationReceived, UnderInvestigation, EvidenceReview, Determination, Sanctioned, Closed

priority	enum		Critical, High, Medium, Low
good_faith_reporting	boolean		Was report made in good faith
standard_of_proof	string		"Preponderance of evidence"
investigator_person_id	UUID	FK	Lead investigator
investigation_start_date	date		When investigation opened
investigation_close_date	date		When closed (NULL if open)
determination	enum		Violation, NoViolation, InsufficientEvidence
violation_severity	enum		Critical, High, Medium, Low
competitive_impact	boolean		Did it affect game outcome
prior_violations_count	integer		Subject's history
cooperation_level	enum		FullCooperation, PartialCooperation, NonCooperation
sanction_type	enum		Fine, Draft Pick, Suspension, Warning, None
sanction_amount	decimal		Financial penalty
sanction_description	text		Full sanction details
historical_comparator	text		Similar past cases for consistency
confidentiality_flag	boolean		TRUE (all cases confidential)
retaliation_concern	boolean		Any retaliation against reporter
case_summary	text		Executive summary

created_at	timestamp	Audit
updated_at	timestamp	Audit

Indexes: case_number, case_status, subject_club_id, domain_id

Security: Highly restricted access, full audit logging

4.7 compliance_incident

Purpose: Non-integrity violations (late submissions, procedural errors, etc.).

Field	Type	Key	Description
incident_id	UUID	PK	Unique identifier
incident_type	enum		LateCertification, MissedMeeting, TechFailure, IncompleteArtifact
domain_id	UUID	FK	Policy domain
clause_id	UUID	FK	Violated clause (if applicable)
incident_date	date		When occurred
club_id	UUID	FK	Responsible club (NULL if League/venue)
game_id	UUID	FK	Related game (NULL if not game-specific)
severity	enum		Critical, High, Medium, Low
description	text		What happened
root_cause	text		Why it happened
corrective_action	text		How it was resolved
resolution_date	date		When resolved
incident_status	enum		Open, InProgress, Resolved, Escalated

escalated_to_case_id	UUID	FK	If escalated to integrity case
responsible_person_id	UUID	FK	Who was accountable
reported_by_person_id	UUID	FK	Who discovered incident
created_at	timestamp		Audit
updated_at	timestamp		Audit

4.8 artifact_submission

Purpose: Tracks submission of required evidence (EMPs, certs, reports, etc.).

Field	Type	Key	Description
submission_id	UUID	PK	Unique identifier
artifact_id	UUID	FK	What was submitted → <code>required_artifact.artifact_id</code>
submitting_entity_type	enum		Club, Venue, Individual
submitting_entity_id	UUID	FK	Who submitted (club_id, person_id, etc.)
season_id	UUID	FK	For which season
game_id	UUID	FK	For which game (NULL if not game-specific)
due_date	date		Policy deadline
submission_date	date		Actual submission
submission_status	enum		OnTime, Late, Pending, Rejected, Approved

days_late	integer		How late (NULL if on time)
approval_status	enum		Pending, Approved, RejectedNeedsRevision
approved_by_person_id	UUID	FK	League approver
approval_date	date		When approved
completeness_score	decimal		0-100% (all required fields present)
file_url	string		Link to submitted document
reviewer_notes	text		Feedback from League
created_at	timestamp		Audit
updated_at	timestamp		Audit

Business Rules:

- Late submissions trigger compliance incidents
- Rejected submissions require resubmission

LAYER 5: METRICS & SCORING

5.1 compliance_score

Purpose: Aggregated compliance rating for clubs/venues/League.

Field	Type	Key	Description
score_id	UUID	PK	Unique identifier
entity_type	enum		Club, Venue, Division, Conference, League
entity_id	UUID	FK	Which entity (club_id, venue_id, etc.)

season_id	UUID	FK	For which season
phase_id	UUID	FK	For which phase (NULL = full season)
domain_id	UUID	FK	For which policy domain (NULL = overall)
score_date	date		When score calculated
overall_score	decimal		0-100 composite score
certification_score	decimal		Sub-score for certs
artifact_submission_score	decimal		Sub-score for required docs
incident_score	decimal		Sub-score (penalized for incidents)
timeliness_score	decimal		Sub-score for meeting deadlines
violation_count	integer		Number of violations in period
incident_count	integer		Number of incidents
rating	enum		Excellent, Good, Fair, NeedsImprovement, Critical
trend	enum		Improving, Stable, Declining
percentile_rank	decimal		vs. other clubs (0-100)
created_at	timestamp		Audit
updated_at	timestamp		Audit

Calculation Logic:

None

```
overall_score = weighted_avg(
    certification_score * 0.30,
    artifact_submission_score * 0.25,
    incident_score * 0.25,
```

```
        timeliness_score * 0.20
    )

    incident_score = 100 - (violation_count * severity_weight)
```

Indexes: `entity_type`, `entity_id`, `season_id`, `domain_id`, `score_date`

5.2 kpi_definition

Purpose: Defines metrics tracked in dashboard.

Field	Type	Key	Description
<code>kpi_id</code>	UUID	PK	Unique identifier
<code>kpi_code</code>	string	UNIQUE	"CERT_COMPLETION_RATE", "EMP_APPROVAL_PCT"
<code>kpi_name</code>	string		"Certification Completion Rate"
<code>description</code>	text		How metric is calculated
<code>domain_id</code>	UUID	FK	Related policy domain (NULL if cross-domain)
<code>unit_of_measure</code>	string		"Percentage", "Days", "Count"
<code>target_value</code>	decimal		Goal (e.g., 100% for cert completion)
<code>warning_threshold</code>	decimal		Yellow alert level (e.g., 90%)
<code>critical_threshold</code>	decimal		Red alert level (e.g., 80%)
<code>calculation_frequency</code>	enum		Daily, Weekly, PerGame, EndOfPhase
<code>data_source_table</code>	string		Which table to query

calculation_logic	text	SQL or formula
created_at	timestamp	Audit
	p	
updated_at	timestamp	Audit
	p	

Example KPIs:

- Certification Completion Rate: % of required certs active
- EMP Approval Rate: % of clubs with approved EMP
- Pregame Meeting Compliance: % of meetings held on time
- Injury Report Timeliness: % of reports submitted on time
- Technology Check Pass Rate: % of tech systems validated

5.3 kpi_snapshot

Purpose: Point-in-time KPI values.

Field	Type	Key	Description
snapshot_id	UUID	PK	Unique identifier
kpi_id	UUID	FK	Which KPI → kpi_definition.kpi_id
entity_type	enum		Club, Division, Conference, League
entity_id	UUID	FK	For which entity
season_id	UUID	FK	For which season
snapshot_date	date		When measured
kpi_value	decimal		Actual value
target_value	decimal		Target at time of snapshot
variance	decimal		Actual - Target
status	enum		OnTarget, Warning, Critical

prior_period_value	decimal	Previous snapshot value
trend	enum	Improving, Stable, Declining
created_at	timestamp	Audit

Indexes: kpi_id, entity_type, entity_id, snapshot_date

5.4 alert_rule

Purpose: Automated compliance alerts and notifications.

Field	Type	Key	Description
alert_rule_id	UUID	PK	Unique identifier
rule_name	string		"Expired Certification Alert"
rule_type	enum		ThresholdBreach, MissedDeadline, AnomalyDetection
trigger_condition	text		Logic for triggering alert
severity	enum		Critical, High, Medium, Low, Info
notification_method	enum[]		Email, SMS, DashboardAlert, APIWebhook
recipient_roles	enum[]		Commissioner, FootballOps, ClubGM, etc.
enabled	boolean		TRUE if active
created_at	timestamp		Audit
updated_at	timestamp		Audit

Example Rules:

- Alert if certification expires in 30 days
 - Alert if pregame meeting delayed >10 minutes
 - Alert if club compliance score drops below 75
 - Alert if injury report late
-

5.5 alert_log

Purpose: History of triggered alerts.

Field	Type	Key	Description
alert_id	UUID	PK	Unique identifier
alert_rule_id	UUID	FK	Which rule triggered
triggered_at	timestamp		When alert fired
entity_type	enum		What entity is affected
entity_id	UUID	FK	Which entity
severity	enum		Critical, High, Medium, Low
alert_message	text		Human-readable message
acknowledged	boolean		Has someone reviewed
acknowledged_by_person_id	UUID	FK	Who acknowledged
acknowledged_at	timestamp		When acknowledged
resolution_status	enum		Open, InProgress, Resolved, Dismissed
resolution_notes	text		How it was handled
created_at	timestamp		Audit

RELATIONSHIP SUMMARY

Key Foreign Key Relationships

None

club → division → conference
club → primary_venue

person → organization (club or league_department)
role_assignment → person, role_definition, club, venue, season

game → home_club, away_club, venue, season
game_role_assignment → game, person, role_definition

certification_record → person, certification_type, season
emergency_management_plan → club, season

injury_report → game, club
uniform_violation → game, club

integrity_case → subject_club, clause, domain
compliance_incident → club, game, clause

compliance_score → entity (club/venue/division), season, domain
kpi_snapshot → kpi_definition, entity

DATA GOVERNANCE

Access Control Matrix

Role	Read Access	Write Access	Delete Access
Commissioner	All data	Integrity cases, sanctions	Archive only
Football Ops	All compliance data	Approvals, incident management	No
Club GM	Own club data only	Submit artifacts, assign roles	No

Game Day Staff	Game-specific data	Meeting logs, tech checks	No
League Observer	Read-only all	No	No
System Admin	All	All	Archive only

Audit Requirements

All write operations logged:

- Who (user_id)
- What (table, record_id, operation)
- When (timestamp)
- Why (business context)
- Where (IP address, device)

Immutable Tables:

- `integrity_case` (append-only)
- `alert_log` (append-only)
- `artifact_submission` (versioned)

Retention Policy:

- Active season data: Hot storage
- Prior 3 seasons: Warm storage
- Archive (4+ years): Cold storage, compliance retention

IMPLEMENTATION NOTES

Database Technology Recommendations

Option 1: PostgreSQL (Relational)

- Strong ACID compliance for audit trail
- JSON fields for flexible policy metadata
- Row-level security for role-based access
- Time-series extensions for KPI snapshots

Option 2: Sentrais Knowledge Graph

- Native entity-relationship modeling
- Graph queries for compliance chains
- Real-time inference (e.g., "find all games with missing certs")
- Temporal versioning built-in

Option 3: Hybrid

- PostgreSQL for transactional data (submissions, incidents)
- Sentrais graph for relationship queries (compliance chains)
- Data warehouse for analytics (Snowflake/BigQuery)

Indexing Strategy

Critical Indexes:

- All foreign keys
- Date fields (game_date, submission_date, expiration_date)
- Status fields (assignment_status, cert_status, incident_status)
- Composite indexes:
 - (club_id, season_id, domain_id) for compliance scoring
 - (game_id, system_type) for tech checks
 - (person_id, cert_type_id, season_id) for cert lookups

API Considerations

RESTful Endpoints:

None

```
GET /api/v1/clubs/{club_id}/compliance-score?season=2024
GET /api/v1/games/{game_id}/pregame-meetings
POST /api/v1/artifact-submissions
GET /api/v1/integrity-cases?status=open
GET /api/v1/dashboard/executive-view?season=2024
```

GraphQL for Complex Queries:

None

```
query ClubReadiness($clubId: UUID!, $seasonId: UUID!) {
```

```
club(id: $clubId) {  
  name  
  complianceScore(seasonId: $seasonId) {  
    overallScore  
    certificationScore  
    rating  
  }  
  roleAssignments(seasonId: $seasonId) {  
    role { name }  
    person { name }  
    certifications { status expirationDate }  
  }  
  emergencyManagementPlan(seasonId: $seasonId) {  
    approvalStatus  
    completenessScore  
  }  
}  
}
```

NEXT STEPS

With this schema foundation, we can now:

1. ☒ Build detailed wireframes for each dashboard view
2. Define specific KPIs and visualizations
3. Create data migration plan from existing NFL systems
4. Develop Sentrais Playbook Translation Engine rules
5. Design role-based UI/UX flows

Schema Version: 1.0

Last Updated: November 15, 2025

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